2024240

I B. E. Examination January – 2024
IT - A , IT -B and All Ex. Engineering
1MERC3 : Elements of Mechanical Engineering

Max. Marks: 60

Time: 3 Hrs

Note: Attempt any **Two Parts** from Each Question. Steam tables are permitted in the examin

1. a. Explain with neat Sketch Quasi static process in thermodynamics 6

   b. A mass of gas is compressed in a quasi static process from 80 kPa , 0.1 m^3 to 0.4 MPa , $0.03m^3$ .Assuming that the pressure and volume are releted by pv^n = constant. Find the work done by the gas system. 6

   c Steam flow steadily through a 0.2 m diameter pipeline from the boiler to the turbine. At the boiler end, the steam conditions are found to be: p = 4 MPa. t = 400 °C. h = 3213.6 kJ/kg, and v = 0.073 m^3/kg. At the turbine end, the conditions are found to be: p = 3.5 MPa, t = 392 °C, h = 3202.6 kJ/kg, and v = 0.084 m3/kg. There is a heat loss of 8.5 kJ/kg from the pipeline. Calculate the steam flow rate 6

2 a. Derive the relation for the efficiency of Otto cycle. 6

  b. In an engine working on Diesel cycle, compression ratio is 16, and at the beginning of compression, pressure and temperature of charge are 1bar and 15 ^{0}C respectively. The temperature at the end of the isobaric process is 1500 ^{0}C. Find : (i) Cut off ratio (ii) Heat supplied per kg of air (iii) cycle efficiency and Mean Effective Pressure 6

  c In a Diesel cycle Cylinder Bore diameter is 15 cm; Stroke length 25 cm; Clearance Volume $400cm^3$. Calculate the efficiency of the cycle if fuel injection takes place at constant pressure for 5% of the stroke. 6

3 a. Ten kg of water at 45^0C is heated at a constant pressure at 10 bar until it becomes superheated steam at 300^0 C Find the change in volume, internal energy, enthalpy, entropy. 6

  b. Steam at 0.8 MPa,250 °C and flowing at the rate of 1 kg/s passes into a pipe carrying wet steam at 0.8 MPa,0.95 dry. After adiabatic mixing, the flow rate is 2.3 kg/s. Determine the condition of steam after mixing. 6

  c Define Dryness fraction of steam. and Explain any one method to Measurement of Dryness fraction of steam. 6

4. a. Define Casting Process. Explain different types of casting defects with their reasons. 6

   b. What are the common Allowances Provided on patterns and Why? 6

   c. Explain different properties of Moulding Sand. Refrac Permea. cost Adn Porosity Cohesive Setting 6

5 a What is welding? How it is classified?. 6

  b Define electric Arc welding. Discuss with the help of neat sketch, the principle of arc welding. 6

  c Name the different types of the lathes available in Machine Shop. Describe the working of a Centre Lathe. 6